

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B.Tech. (IT) Examination

June 2022

IT254 Computer Architecture & Microprocessor Interfacing

Date: 27.06.2022, Monday.

Time: 10.00 a.m. To 01.00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.

SECTION – I

- Q - 1 Answer the Following Questions. [06]**
- a. If clock rate is 50 MHZ, find execution time for a program with 1000 instructions, if CPI for program is 3.5. [3]
 - b. What will be the content of the register X11 after the execution of the following code? [3]
1) ADDI X11, \$ZERO, #202. 2) SLL X19, X11, 6. 3) ADD X11, X11, X38
- Q - 2 Answer the Following Questions. (Any Three) [15]**
- a. What is Embedded Computer? Explain with the Suitable Examples. [5]
 - b. 50 billion instructions are given for the 4 different tasks. 5 billion instructions are for Branches (CPI = 4), 15 billion instructions are for Loads (CPI = 2), 10 billion instructions are for Store (CPI = 3) and rest instructions are for integer add (CPI = 1). If clock rate = 4 GHz then what will be the execution time? [5]
 - c. Explain Hardware Abstraction. [5]
 - d. Convert following code to equivalent MIPS code: [5]
a = a OR b
b = b AND 5
c = NOT a
- Q - 3 Answer the Following Questions. (Any Two) [14]**
- a. Explain the Various Addressing Modes Followed in MIPS. [07]
 - b. Explain Various Arithmetic and Logical Instructions in MIPS. [07]
 - c. Explain PowerPC Architecture. [07]

SECTION – II

- Q - 4 Answer the Following Questions. [06]**
- a. Convert the below IEEE754 floating point number to decimal. [3]
0 01111011 01011100001010001111011
 - b. Explain the use of Program Counter and Instruction Register. [3]
- Q - 5 Answer the Following Questions. (Any Three) [15]**
- a. Explain the Four-Segment Instruction Pipeline with the suitable diagram. [5]
 - b. Write the difference between RISC and CISC. [5]
 - c. Explain Direct Mapped Cache. [5]
 - d. What are Structural Hazards and Branch Hazards? How These Hazards could be Solved? [5]
- Q - 6 Answer the Following Questions. (Any Two) [14]**
- a. Explain IEEE 754 Format using Suitable Example. [07]
 - b. Compute (10111) X (10011) using Booth Algorithm. The given data is represented in Signed 2's Complement. [07]
 - c. Draw the Single Cycle Data Path for "ADD" Instruction. [07]